Assignment: Software Testing and Quality Assurance (MCA 2nd YEAR)

1. What is Software Testing? Explain its objectives and importance.
2. What is Software Quality Assurance (SQA)? Explain its role in software development.
3. Differentiate between Software Testing and Quality Assurance.
4. Explain the major objectives of Software Quality Assurance.
5. What is the difference between Quality Assurance (QA) and Quality Control (QC)?
6. Explain the role and responsibilities of a Software Quality Assurance Engineer.
7. Explain the relationship between Software Testing, Quality Assurance, and Quality Control.
8. What are the different activities performed during Software Quality Assurance?
9. Explain how Software Testing helps in improving software quality.
10. What is a Quality Management System (QMS)? Explain its importance in software development.
11. Explain the importance of standards, processes, and documentation in Software Quality Assurance.
12. What is a Software Quality Assurance Plan (SQAP)? Explain its major components.
13. Explain the difference between Verification and Validation and discuss their role in software quality.
14. Explain how defects can affect the overall quality of software.
15. A banking application frequently crashes during online transactions. As a QA professional,
16. explain what activities you would perform to improve the quality of the application.
17. Explain the importance of continuous improvement in Software Quality Assurance.
18. What are the major challenges faced while implementing Software Quality Assurance in an organization?

19. Explain the role of reviews, inspections, and audits in Software Quality Assurance.

20. Explain how QA activities are performed throughout the Software Development Life Cycle (SDLC).

21. Explain with a suitable real-world example how effective Software Testing and Quality Assurance can improve customer satisfaction.